

CERTIFICATE

This is to certify that the Main - Project entitled

CAREER PATH

Being submitted to

St. Joseph's College of Arts and Science (Autonomous),

(Affiliated to Annamalai University – Annamalai Nagar)

Submitted By

A. AROCKIA JOHN JOSEPH [Reg No. A24MCS01]

For the partial fulfillment for the award of degree of

MASTER OF SCIENCE IN COMPUTER SCIENCE

Is a Bonafide record of work carried out by them,

Under my guidance and supervision.

INTERNAL GUIDE

HEAD OF THE DEPARTMENT

Submitted for the viva-voce examination held on -----

Examiners:

1. -----

2. -----

ABSTRACT

The "**Career Path**" is an advanced Student Information System (SIS) designed to transform traditional academic record-keeping into a data-driven career forecasting tool. Current manual systems often fail to provide students with a clear understanding of their academic growth or professional direction. This project addresses that gap by implementing a centralized platform built on a Python (Flask) backend and a MySQL database, ensuring secure and scalable data management with a focus on data integrity.

The system features a "**Neural Vault**" interface that provides high-fidelity visualizations of semester-wise performance, GPA trends, and attendance through interactive trajectory charts. A key functional highlight is the automated evaluation engine that processes raw academic data against predefined performance benchmarks to generate personalized career recommendations. By analysing subject-specific strengths and historical academic patterns, the platform acts as a strategic bridge between current studies and future professional goals. Ultimately, "Career Path" delivers a visually polished, mobile-responsive solution that empowers students with actionable insights, making academic tracking more intuitive, analytical, and goal-oriented.

ACKNOWLEDGEMENT

It is my earnest and sincere desire and ambition to acquire profound knowledge in the study of **Master of Science in Computer Science**. I grateful to God, the Almighty who has blessed us abundantly and guided us to complete this task.

I express my sincere gratitude to our beloved, **Rev. Fr. Dr. M. Swaminathan, Secretary**, St. Joseph's College of Arts and Science (Autonomous), Cuddalore, for providing such a good congenial environment to enlighten our knowledge.

I express my sincere gratitude to our respected **Dr. M. Arumai Selvam, M.Sc., M.Phil., Ph.D., Principal and Head, Post Graduate and Research Department of Computer Science & Artificial Intelligence**, St. Joseph's College of Arts and Science (Autonomous), Cuddalore, for his support and encouragement.

I take immense pleasure in conveying my sincere and heartfelt gratitude to our internal guide **Dr. M. Arumai Selvam, M.Sc. (Maths)., M.Sc. (CS)., M.Phil. (CS)., Ph.D., PGDCSA., Principal and Head, Post Graduate and Research Department of Computer Science and Artificial Intelligence** for her valuable guidance, inspiration, consent and motivation throughout my project. her untiring encouragement made my thoughts enthusiastic in successfully completing this project.

At the Outset, I extend my sincere gratitude to all our teachers in the Department of Computer Science department who have taken pains to shape and mold us in all my endeavors. I will fail in our duty if i do not put on record my heartfelt gratitude to my beloved parents for their financial and material support as well their words of encouragement and motivation to complete this project successfully.

Last but not the Least, we place my deepest sense of gratitude to all my Friends who had inspired and supported me to complete this project more successfully.

A. AROCKIA JOHN JOSEPH
[A24MCSC01]

TABLE OF CONTENTS

CERTIFICATE	i
ACKNOWLEDGEMENT	ii
ABSTRACT	iii

SL. NO	CONTENT	PAGE NO.
1.	INTRODUCTION 1.1 ABOUT THE PROJECT	01
2.	PROBLEM DEFINITION 2.1 EXISTING SYSTEM 2.2 DISADVANTAGES OF EXISTING SYSTEM 2.3 PROPOSED SYSTEM 2.4 ADVANTAGES OF PROPOSED SYSTEM	02
3.	SOFTWARE REQUIREMENTS SPECIFICATION 3.1 INTRODUCTION 3.2 MODULE DESCRIPTION 3.3 HARDWARE REQUIREMENTS 3.4 SOFTWARE REQUIREMENTS	04
4.	SYSTEM ANALYSIS 4.1 INTRODUCTION 4.2 SEQUENCE DIAGRAM 4.3 DATA FLOW DIAGRAM	06

5	SYSTEM DESIGN 5.1 INTRODUCTION 5.2 USER INTERFACE DESIGN	08
6	SYSTEM TESTING 6.1 INTRODUCTION 6.2 UNIT TESTING 6.3 INTEGRATION TESTING 6.4 VALIDATION TESTING 6.5 ALPHA AND BETA TESTING 6.6 SYSTEM TESTING	10
7	CONCLUSION 7.1 CONCLUSION 7.2 FUTURE ENHANCEMENTS	12
8	BIBLIOGRAPHY 8.1 BOOKS 8.2 WEBSITE	13
	APPENDICES APPENDIX-A DESCRIPTION APPENDIX-B SAMPLE SOURCE CODE APPENDIX-C SCREEN SHOTS	21